

Sr₂PrSbO₆ Double Perovskite as a Solar-Blind UV Photodetector: A Combined DFT and Device Simulation Study

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Abstract

The monoclinic double perovskite oxide Sr₂PrSbO₆ (space group P2₁/c, No. 14) crystallises with rock-salt ordering of Pr³⁺ and Sb⁵⁺ on the B-sublattice, driven by the 3+/5+ charge difference of the rare-earth and pnictogen cations. First-principles calculations with the screened hybrid HSE06 functional yield an indirect band gap of 5.090 eV, which fixes an intrinsic optical absorption cutoff at $\lambda_c = \frac{hc}{E_g} = 243.6$ nm, placing Sr₂PrSbO₆ within the solar-blind ultraviolet (UV-C) window (200–280 nm). Coupling the HSE06 electronic structure to a self-consistent drift-diffusion (TCAD) simulation of a metal–semiconductor–metal photoconductor with platinum Schottky contacts shows that the responsivity vanishes for all wavelengths above 243.6 nm, giving an effectively infinite UV/visible rejection ratio together with an ultralow dark current and a photocurrent contrast exceeding 10¹³ at 10 V. The HSE06 optical data yield $\alpha(220 \text{ nm}) = 5.02 \times 10^4 \text{ cm}^{-1}$ from direct-allowed transitions above the indirect gap, giving $\eta_{\text{abs}} \approx 53\%$ at $d = 200$ nm; the baseline device achieves $R = 2.82 \text{ mA/W}$, $D^* = 1.6 \times 10^9$ Jones, and nanosecond response, with performance governed by carrier transport rather than optical absorption. The thermodynamic, mechanical, and dynamic stability of Sr₂PrSbO₆ identify it as a wide-gap oxide candidate for deep solar-blind UV-C photodetection.

Keywords: *Double perovskite; HSE06 hybrid functional; Wide band gap; Solar-blind UV-C photodetector; Pr-4f localisation; First principles*

1. Introduction

The solar-blind UV window (200–280 nm, UV-C) is completely extinguished at ground level by stratospheric ozone and molecular oxygen, so any photon detected in this band must originate from a terrestrial source. Photodetectors that respond exclusively within this window therefore operate against an intrinsically zero solar background, enabling high-contrast, low-false-alarm detection for missile plume tracking, flame and corona-discharge sensing, non-line-of-sight secure optical communication, ozone monitoring, and biochemical analysis^{1–3}. Realising true solar blindness requires a semiconductor with a band gap exceeding ~4.43 eV ($\lambda < 280$ nm), and the past two decades have produced an intensively studied portfolio of ultrawide band gap candidates: β -Ga₂O₃ ($E_g \approx 4.8$ eV), the material most closely identified with intrinsic solar-blind response^{4,5}; high-Al-content Al_xGa_{1-x}N alloys with gaps tunable from 3.4 to 6.2 eV, which have advanced to focal-plane-array imaging^{6,7}; and MgZnO and diamond^{8–10}. Each platform nonetheless carries well-documented limitations, such as phase instability and polymorphism in

Ga₂O₃, severe epitaxial and p-type doping difficulties in high-Al AlGa_{1-x}N, phase segregation in MgZnO, and the cost and material immaturity of diamond.^{9,11} These limitations motivate the search for alternative ultrawide-gap semiconductors, particularly chemically stable oxides, that combine a detection edge deeper in the UV-C than the ~259 nm cutoff of β -Ga₂O₃ with a single, stable, non-polymorphic phase, mechanical robustness, and adequate thermal conductivity.

Perovskite-structured oxides and halides have recently emerged as a credible third platform for UV photodetection. Two-dimensional perovskite oxide Sr₂Nb₃O₁₀ nanosheets deliver a responsivity of 1214 A W⁻¹ and detectivity of 1.4×10^{14} Jones at 270 nm¹², the lead-free halide double perovskite Cs₂AgBiCl₆ has been deposited as films for highly selective and stable UV photodetectors⁷, and metal–semiconductor–metal (MSM) β -Ga₂O₃ devices with graphene electrodes set the responsivity benchmark for the architecture (≈ 29.8 A W⁻¹ at 258 nm) against which new materials are judged¹³. More broadly, the perovskite motif offers band gaps that are tunable through A/B-site chemistry and dimensionality, low-temperature and solution-based processing routes, and, in the halides, a pronounced defect tolerance^{14,15}, which is why perovskite oxides and halides are now being explored across the entire ultraviolet range, including self-powered double-perovskite-oxide photodetectors¹⁶. These demonstrations establish that the perovskite structural motif is fully compatible with high-performance UV detection. However, the double perovskite *oxides* A₂BB'O₆ remain relatively unexplored for solar-blind detection, despite hosting some of the widest, most thermally and chemically stable gaps in the oxide family, and no member of the family has been carried from first-principles electronic structure through to predicted device-level figures of merit.

Sr₂PrSbO₆ belongs to the rare-earth B-site-ordered double perovskites A₂LnB'O₆ (A = Ca, Sr, Ba; B' = Sb⁵⁺, Nb⁵⁺, Ta⁵⁺, Bi⁵⁺, W⁶⁺), in which the 3+/5+ charge contrast between Ln³⁺ and B' drives complete rock-salt ordering of the B sublattice^{17,18}. The family has been developed mainly as luminescent hosts—Eu³⁺-activated Ca₂LnSbO₆ and Sr₂LnSbO₆ phosphors and luminescence thermometers^{19–21}—as low-loss microwave dielectrics²², as frustrated-lattice magnets and magnetocalorics^{17,23}, and more recently as photosensitive and photocatalytic oxides²⁴; a self-powered photodetector has been demonstrated on the related Bi/Nb double perovskite¹⁶, but no photodetector based on a lanthanide-containing A₂LnB'O₆ oxide has been reported. First-principles studies of the cubic or near-cubic Ba₂LnSbO₆ and Ba₂LnTaO₆ compounds accordingly find direct Γ -point gaps formed by O-2p $\rightarrow$ Sb-5s/Ln-5d (or Ta-5d) transitions^{24–27}, and the nearly cubic Pr compound Ba₂PrWO₆ was recently shown to be an insulator with a gap above 4.93 eV and mixed Pr³⁺/Pr⁴⁺ valence²⁸. For Pr-containing perovskites the computed electronic structure also depends strongly on the treatment of the Pr 4f shell: spin-polarised GGA(+U) calculations that place the 4f² states at the Fermi level predict half-metallic Sr₂PrSnO₆ and PrAlO₃^{29,30}, whereas treatments that keep the localised 4f² core of Pr³⁺ away from the band edges—4f-in-core pseudopotentials^{31,32} or screened hybrid functionals³³—yield wide-gap insulators whose conduction bands are Pr-5d/B'-derived, as found for PrAlO₃ and PrGaO₃³⁴ and for Cs₂NaPrCl₆-type halide double perovskites³⁵. Despite this growing body of work, neither a hybrid-functional characterisation of Sr₂PrSbO₆ nor any assessment of its deep-UV optoelectronic potential let alone a first-principles-to-device evaluation has previously appeared in the peer-reviewed literature. In addition to that, relative to established wide-band-gap UV photodetector platforms, Sr₂PrSbO₆ presents several potentially attractive characteristics that motivate its investigation as an alternative UV-absorbing material. The wide-gap electronic structure of related Sr₂XSbO₆ double perovskites³⁶ suggests the possibility of an intrinsic deep-UV absorption edge without relying on alloying strategies, thereby avoiding composition-control issues associated with high-Al Al_xGa_{1-x}N and phase-segregation concerns reported for MgZnO. This considerations provide a strong motivation for systematically evaluating Sr₂PrSbO₆ from both the materials and device perspectives.

In this work we close that gap through a multiscale first-principles investigation of $\text{Sr}_2\text{PrSbO}_6$ that proceeds from parameter-free DFT calculation to device implication. We address the following interconnected questions: (i) Does the compound simultaneously satisfy the thermodynamic, mechanical, and dynamic stability criteria that underpin experimental synthesizability? (ii) What is the correct HSE06 electronic structure, which orbitals define the band edges, and what does it imply for intrinsic dopability? (iii) What solar-blind photodetector performance does this electronic structure imply at the device level? To answer them we couple the HSE06 band gap and optical absorption spectrum to self-consistent TCAD drift-diffusion simulations (DEVSIM³⁷) of a Pt/ $\text{Sr}_2\text{PrSbO}_6$ /Pt MSM photoconductor, extracting spectral responsivity, external quantum efficiency, specific detectivity, transient response, and UV/visible rejection.

2. Computational Details

2.1. Code and Pseudopotentials

All density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP)^{38,39} within the projector augmented-wave (PAW) formalism^{40,41}. The valence electron configurations treated explicitly are: Sr ($4s^2 4p^6 5s^2$), Pr ($5s^2 5p^6 6s^2$), Sb ($5s^2 5p^3$), and O ($2s^2 2p^4$). The plane-wave kinetic energy cutoff was set to 600 eV, and the Brillouin zone was sampled with a $7 \times 6 \times 4$ Monkhorst–Pack mesh after a convergence test. The GGA-PBE calculations⁴² (elastic constants, equation of state, formation energy) used a 450 eV cutoff with $7 \times 6 \times 4$ Monkhorst–Pack mesh.

2.2. Structural Relaxation

Full structural relaxation of atomic positions, cell shape, and cell volume was carried out using the screened hybrid HSE06 functional^{43,44}. The conjugate-gradient algorithm was used for ionic updates. All calculations were performed without spin polarisation.

2.3. Electronic Band Structure and Density of States

The electronic band structure and projected density of states (PDOS) were computed with the HSE06 functional in non-self-consistent-field (NSCF) mode. The k-path for the band structure was generated by VASPKIT⁴⁵. All calculations in this work were performed without spin polarisation.

2.3.1 Branch-point energy

To assess the intrinsic dopability of $\text{Sr}_2\text{PrSbO}_6$, the branch-point energy (charge-neutrality level) E_{BP} was evaluated from the HSE06 eigenvalues as the Brillouin-zone average of the mid-point between the N_{CB} lowest conduction bands and the N_{VB} highest valence bands, $E_{\text{BP}} = (1/2N_k) \sum_k [(1/N_{\text{CB}}) \sum_i \epsilon_{c,i}(\mathbf{k}) + (1/N_{\text{VB}}) \sum_j \epsilon_{v,j}(\mathbf{k})]$, following Tersoff⁴⁶ and Schleife et al.⁴⁷ The average was taken over the 24 symmetry-irreducible points of the Γ -centred $5 \times 4 \times 3$ mesh with their weights; $N_{\text{CB}} = 2$ and $N_{\text{VB}} = 4$ were used for the two-formula-unit cell, and the sensitivity to this choice was checked (Table S16).

2.4. Lattice Dynamics and Phonon Properties

Harmonic interatomic force constants (IFCs) were computed using the finite-displacement approach implemented in Phonopy v2.42⁴⁸ interfaced with VASP. Non-analytic corrections (NAC) were applied at the Γ point to recover the LO-TO splitting of infrared-active modes. Thermodynamic quantities such as the vibrational heat capacity C_v , entropy S , Helmholtz free energy F , and vibrational energy E_{vib} were extracted as a function of temperature (0–1000 K) using the Phonopy post-processing module.

2.5. Elastic Constants and Mechanical Properties

The full elastic stiffness tensor C_{ij} was determined by the finite-difference method within the GGA-PBE functional⁴². Post-processing to derive polycrystalline Voigt–Reuss–Hill (VRH) averages bulk modulus B , shear modulus G , Young's modulus E , and Poisson's ratio ν was performed with VASPKIT⁴⁵.

2.6 Formation Enthalpy and Thermodynamic Functions

The formation enthalpy was evaluated at $T = 0\text{K}$ in the static=lattice approximation, $\Delta H_f = E(\text{Sr}_2\text{PrSbO}_6) - [2E(\text{Sr}) + E(\text{Pr}) + E(\text{Sb}) + 3E(\text{O}_2)]$, from the GGA-PBE total energies computed with a 600 eV cutoff and identical PAW potentials for the compound and the elemental ground states. Because PBE over-binds the O_2 molecule.

2.7. Carrier Mobility

Carrier mobilities were estimated within Bardeen–Shockley deformation-potential theory⁴⁹. The band-edge effective masses m^* were obtained by parabolic fitting of the HSE06 conduction- and valence-band extrema, the elastic modulus C from the computed stiffness tensor, and the deformation-potential constant $E_1 = \partial E_{\text{edge}} / \partial \epsilon$ from the linear shift of the CBM (electrons) and VBM (holes) under uniaxial strains of $\pm 1.0\%$. The room-temperature carrier mobility was then evaluated at $T = 300\text{ K}$ as: $\mu = [2\sqrt{(2\pi)} \cdot e \cdot \hbar^4 \cdot C] / [3 \cdot (m^*)^{5/2} \cdot (k_B \cdot T)^{3/2} \cdot E_1^2]$ where e is the elementary charge, $\hbar$ the reduced Planck constant, k_B the Boltzmann constant, and T the temperature.

2.8. Device Simulation

Device-level simulations were performed using the open-source TCAD software DEVSIM (version 2.10.0),³⁷ which solves the self-consistent system of Poisson's equation and the electron/hole drift-diffusion continuity equations on an unstructured one-dimensional finite volume mesh. The Scharfetter–Gummel scheme was used for carrier current discretization, ensuring numerical stability under high electric fields.

The mesh extends from $x = 0$ (anode contact) to $x = L = 10\text{ }\mu\text{m}$ (cathode), with node spacing refined to 5 nm near both contacts to resolve the depletion regions, and relaxed to 50 nm in the bulk. Shockley–Read–Hall (SRH) recombination was included using carrier lifetimes $\tau_n = \tau_p = 1\text{ ns}$, consistent with the nanosecond-scale photoexcited-carrier dynamics measured in wide-bandgap oxide solar-blind photodetector materials such as $\beta\text{-Ga}_2\text{O}_3$.⁵⁰ Optical generation was incorporated as a uniform volumetric source term $G = \eta_{\text{abs}} \times \Phi_0 / d$, where $\eta_{\text{abs}} = (1 - R)(1 - e^{-\alpha d})$ is the Beer–Lambert film absorption efficiency, R is the front-surface reflectance, α is the

wavelength-dependent absorption coefficient, d is the film thickness, and Φ_0 is the incident photon flux.

Schottky contact boundary conditions were implemented using the standard thermionic emission. At the two Pt contacts the electrostatic potential is fixed to the applied bias and the carrier densities are fixed at their equilibrium values (Dirichlet boundary conditions), the standard DEVSIM contact model. The Pt⁵¹ work function ($\Phi_M = 5.65$ eV) enters through the band alignment: because the film is undoped ($n_i \approx 10^{-24}$ cm⁻³), its Fermi level is fixed by the contacts at $E_F = -\Phi_M$ throughout the gap (Section 3.5.1), giving barriers $\Phi_{B,n} = \Phi_M - \chi = 2.15$ eV and $\Phi_{B,p} = E_g - \Phi_{B,n} = 2.94$ eV. Thermionic emission over these barriers, $J = A^* T^2 \exp(-\Phi_B/k_B T) \approx 4 \times 10^{-30}$ A cm⁻² ($A^* \approx 60$ A cm⁻² K⁻² for $m_e^* = 0.5 m_0$), corresponds to $\approx 10^{-35}$ A for the $100 \mu\text{m} \times 200 \text{ nm}$ contact cross-section and is therefore not resolved by the simulation; the simulated dark current ($< 10^{-24}$ A) is set by the numerical carrier-density floor of 10^{-3} cm⁻³ and should be read as an upper bound. For D^* calculations, a practical dark current density of $J_{\text{dark}} = 10^{-10}$ A/cm² was adopted from experimentally reported values for comparable wide-bandgap MSM Schottky photodetectors¹³. This conservative estimate avoids the non-physical D^* values that would arise from assuming purely thermionic dark current over a 2.15 eV barrier at 300 K. A complete summary of all simulation parameters, including DFT-derived, literature-sourced, and estimated quantities, is provided in supplementary Table S9.

The coupled system solved at each bias point consists of the electrostatic Poisson equation and the electron and hole continuity equations under steady-state (DC) conditions⁵²:

$$\nabla \cdot (\epsilon \nabla \phi) = -q(p - n - N_D - N_A) \quad (1)$$

$$\nabla \cdot J_n = qU_{\text{SRH}} - qG_{\text{opt}} \quad (2a)$$

$$\nabla \cdot J_p = -qU_{\text{SRH}} + G_{\text{opt}} \quad (2b)$$

where ϕ is the electrostatic potential, $\epsilon = \epsilon_r/\epsilon_0$ is the permittivity, q is the elementary charge, n and p are the free electron and hole concentrations, N_D and N_A are the ionised donor and acceptor densities (intrinsic, undoped material), J_n and J_p are the electron and hole current densities, U_{SRH} is the net Shockley–Read–Hall (SRH) recombination rate, and G_{opt} is the optical carrier generation rate.

The drift-diffusion current densities are⁵²:

$$J_n = q\mu_n nE + qD_n \nabla n \quad (3a)$$

$$J_p = q\mu_p pE + qD_p \nabla p \quad (3b)$$

where μ_n and μ_p are the electron and hole mobilities, $D_{\{n,p\}} = \mu_{\{n,p\}} k_B T / q$ are the corresponding diffusion coefficients (Einstein relation), and $E = -\nabla \phi$ is the local electric field.

Trap-mediated recombination is modelled using the standard SRH expression^{53,54} :

$$U_{SRH} = (np - ni^2) / [\tau_p(n + n_1) + \tau_n(p + p_1)] \quad (4)$$

The simulated device is a planar metal–semiconductor–metal (MSM) photoconductor comprising two Pt contacts separated by a gap of $L = 10 \mu\text{m}$ across an undoped $\text{Sr}_2\text{PrSbO}_6$ film of thickness $d = 200 \text{ nm}$ and nominal width $W = 100 \mu\text{m}$. The 1D mesh spans the inter-electrode gap ($0 \leq x \leq L$) and is non-uniform: node spacing is 5 nm within $\sim 100 \text{ nm}$ of each contact (to resolve the near-contact depletion/accumulation region) and increases to $\sim 50 \text{ nm}$ in the quasi-neutral bulk.

Fixed-potential boundary conditions are applied at both Pt contacts: the electrostatic potential is set to the applied bias (anode) or ground (cathode), and the carrier quasi-Fermi levels are pinned at their equilibrium values. No explicit Schottky thermionic-emission current is imposed; the blocking character emerges naturally from the large band gap and the pinned contact quasi-Fermi levels.

The simulation follows a two-step convergence protocol. First, only the Poisson equation is solved with carrier densities fixed at their intrinsic thermal-equilibrium values to obtain a physically consistent initial electrostatic potential. Second, the full coupled drift-diffusion system (Eqs. 1–3) is activated and solved self-consistently using Newton's method until relative residuals fall below 10^{-3} (carrier equations) and 10^{-10} (Poisson). DC bias sweeps are performed by incrementally stepping the anode potential from 0 to 20 V in 0.5 V steps, using the converged solution at each step as the initial guess for the next. Dark ($G_{\text{opt}} = 0$) and illuminated ($G_{\text{opt}} > 0$) sweeps are computed sequentially in order to avoid convergence difficulties associated with transitioning from a high-carrier-density photo-excited state back to a near-intrinsic dark state.

Following the FV-DD simulations, spectral and bias-dependent performance metrics are computed analytically from the photoconductor gain model⁵²:

$$G_{pc} = \tau(\mu_n + \mu_p)V/L^2 \quad (7)$$

$$R(\lambda) = \eta_{\text{abs}}(\lambda) \times G_{pc} \times q / (h\nu) \quad (8)$$

$$EQE(\lambda) = R(\lambda) \times \frac{h\nu}{q} \times 100\% \quad (9)$$

$$D^*(\lambda) = R(\lambda) \sqrt{A_{\text{det}}} / \sqrt{(2qJ_{\text{dark}})} \quad (10)$$

where G_{pc} is the photoconductor gain, V is the applied bias, L is the electrode gap, $A_{\text{det}} = L \times W$ is the detector active area, and J_{dark} is the dark current density.

To visualise the two-dimensional spatial distribution of the electric potential, carrier densities, and optical generation across both the thickness (z -axis) and the lateral gap (x -axis) of the device, a 2D finite-volume simulation was also performed on a structured rectangular mesh using the same drift-diffusion equations and material parameters. The 2D results, including spatial maps of E-field, electron/hole concentration, and generation rate, are presented in the supplementary material and are fully consistent with the 1D results reported in the main text.

3. Results and Discussion

3.1. Crystal Structure and Structural Parameters

$\text{Sr}_2\text{PrSbO}_6$ adopts the monoclinic $\text{P2}_1/\text{c}$ (No. 14) space group with two formula units per conventional cell. The HSE06-relaxed lattice parameters are $a = 5.887 \text{ \AA}$, $b = 6.004 \text{ \AA}$, $c = 10.214 \text{ \AA}$, and $\beta = 124.82^\circ$, giving a unit cell volume of $V = 296.38 \text{ \AA}^3$ ($148.19 \text{ \AA}^3$ per formula unit). The Goldschmidt tolerance factor, evaluated using Shannon ionic radii⁵⁵ for twelvefold-coordinated Sr^{2+} ($1.44 \text{ \AA}$), sixfold-coordinated Pr^{3+} ($0.99 \text{ \AA}$) and Sb^{5+} ($0.60 \text{ \AA}$), and O^{2-} ($1.40 \text{ \AA}$), is $t = 0.915$. This value lies within the 0.89–1.00 range characteristic of tilted perovskite structures⁵⁶.

The crystal structure (Fig. 1) consists of corner-sharing PrO_6 and SbO_6 octahedra arranged in a rock-salt ordering pattern on the B-sublattice, with Sr^{2+} ions occupying the A-site cavities between the octahedra. The PrO_6 octahedra are slightly elongated and distorted: the six Pr–O bond lengths span 2.335–2.355 $\text{\AA}$ (mean 2.346 $\text{\AA}$). The SbO_6 octahedra are considerably more regular, with Sb–O distances in the range 2.006–2.016 $\text{\AA}$ (mean 2.011 $\text{\AA}$), consistent with the compact, highly charged Sb^{5+} cation forming strong Sb–O bonds. The optimised structural parameters are collected in Table 1.

Table 1. Optimised structural parameters of $\text{Sr}_2\text{PrSbO}_6$ from HSE06 calculations.

Parameter	Value	Notes	Experimental ⁵⁷
Formula	$\text{Sr}_2\text{PrSbO}_6$	Double perovskite	
Space group	$\text{P2}_1/\text{c}$ (No. 14)	Monoclinic	$\text{P2}_1/\text{c}$ (No. 14)
Crystal system	Monoclinic	Point group 2/m	Monoclinic
Z (f.u. per cell)	2	20 atoms per cell	
a ($\text{\AA}$)	5.887	HSE06-relaxed	5.87905
b ($\text{\AA}$)	6.004		5.96952
c ($\text{\AA}$)	10.214		8.36920
β ($^\circ$)	124.82		
V ($\text{\AA}^3$)	296.38	$148.19 \text{ \AA}^3$ per f.u.	293.71
Goldschmidt factor t	0.915	Using Shannon ionic radii	
Pr–O bond length ($\text{\AA}$)	2.335–2.355	mean 2.346 $\text{\AA}$	

Parameter	Value	Notes	Experimental ⁵⁷
Sb–O bond length (Å)	2.006–2.016	mean 2.011 Å	
Sr–O bond length (Å)	2.50–2.99	8 nearest O neighbours	

The energy–volume curve fitted to the Murnaghan equation of state is shown in Fig. 1. The Murnaghan EOS parameters are summarised in Supplementary Table S10. The GGA-PBE equilibrium volume $V_0 = 15.894 \text{ Å}^3/\text{atom}$ (317.9 Å^3 per conventional cell) is approximately 7% larger than the HSE06-relaxed volume (296.4 Å^3). The bulk modulus $B_0 = 125.5 \text{ GPa}$ from the EOS is in good agreement with $B_H = 121.6 \text{ GPa}$ from the elastic tensor (Table 2), with the 3% deviation within the expected range for the two methods applied at slightly different volumes.

3.2. Stability Analysis

Predicting synthesisability requires that a candidate compound satisfy three independent stability criteria^{58,59}: thermodynamic stability against decomposition into competing phases, mechanical stability against elastic deformation, and dynamic stability against lattice phonon collapse.

3.2.1. Thermodynamic Stability

The formation enthalpy relative to the elemental ground states (fcc Sr, dhcp Pr, A7 Sb, O₂; Eq. 11) computed at 0 K in the static-lattice approximation is $\Delta H_f = -2.61 \text{ eV/atom}$ (-26.1 eV per formula unit, $-2518 \text{ kJ mol}^{-1}$); the zero-point correction is $+0.02 \text{ eV/atom}$, and applying the empirical O₂ over-binding correction^{60,61} gives -3.02 eV/atom . These values bracket the sum of the experimental formation enthalpies of the constituent binary oxides, $2 \text{ SrO} + \frac{1}{2} \text{ Pr}_2\text{O}_3 + \frac{1}{2} \text{ Sb}_2\text{O}_5 = -2.67 \text{ eV/atom}$,^{62,63} which is a lower bound on $|\Delta H_f|$ of the double perovskite. $\text{Sr}_2\text{PrSbO}_6$ is therefore strongly stable against decomposition into its elements. Its thermodynamic stability is in fact established experimentally: the compound has been synthesised by high-temperature solid-state reaction and its $\text{P2}_1/\text{n}$ structure refined by Wong-Ng et al.,⁵⁷ and the HSE06 lattice parameters of Table 1 reproduce the experimental cell to within 0.1 % (a), 0.6 % (b), 0.2 % (c) and 0.9 % (volume).

The temperature dependence of the stability is shown in Fig. S17. The vibrational entropy of $\text{Sr}_2\text{PrSbO}_6$ rises to $25.0 \text{ J K}^{-1} \text{ mol}^{-1} \text{ atom}^{-1}$ at 300 K and the heat capacity reaches 98 % of the Dulong–Petit limit at 1000 K. The formation entropy is dominated by the loss of the translational entropy of three O₂ molecules, $\Delta S_f \approx -0.52$ to $-0.56 \text{ kJ K}^{-1} \text{ mol}^{-1}$ per formula unit, so that the Gibbs free energy ΔG_f becomes less negative with temperature; nevertheless ΔG_f remains -2.43 eV/atom at 300 K and -2.04 eV/atom at 1000 K (-2.84 and -2.45 eV/atom with the O₂ correction), i.e. the compound is thermodynamically stable over the entire range relevant to device operation and to ceramic processing. Consistent with this, the isostructural antimonates $\text{Sr}_2\text{LnSbO}_6$ are prepared in air at high temperature,^{64,65} and the $\text{Sr}_2\text{SmSbO}_6$ and $\text{Sr}_2\text{LaSbO}_6$ end members undergo only displacive octahedral-tilt transitions, $\text{P2}_1/\text{n} \rightarrow \text{R-3}$ (885 and 945 K) $\rightarrow \text{Fm-3m}$ (1170 and 1180 K), without decomposition;⁶⁴ Pr^{3+} lies between Sm^{3+} and La^{3+} in ionic radius, so $\text{Sr}_2\text{PrSbO}_6$ is expected to follow the same sequence at $\approx 900\text{--}950 \text{ K}$ and $\approx 1175 \text{ K}$. The monoclinic $\text{P2}_1/\text{c}$ phase treated here is thus the stable low-temperature polymorph, and the perovskite framework persists to the synthesis temperature; the phonon spectrum of Section 3.2.3 confirms that it is a true minimum at 0 K.

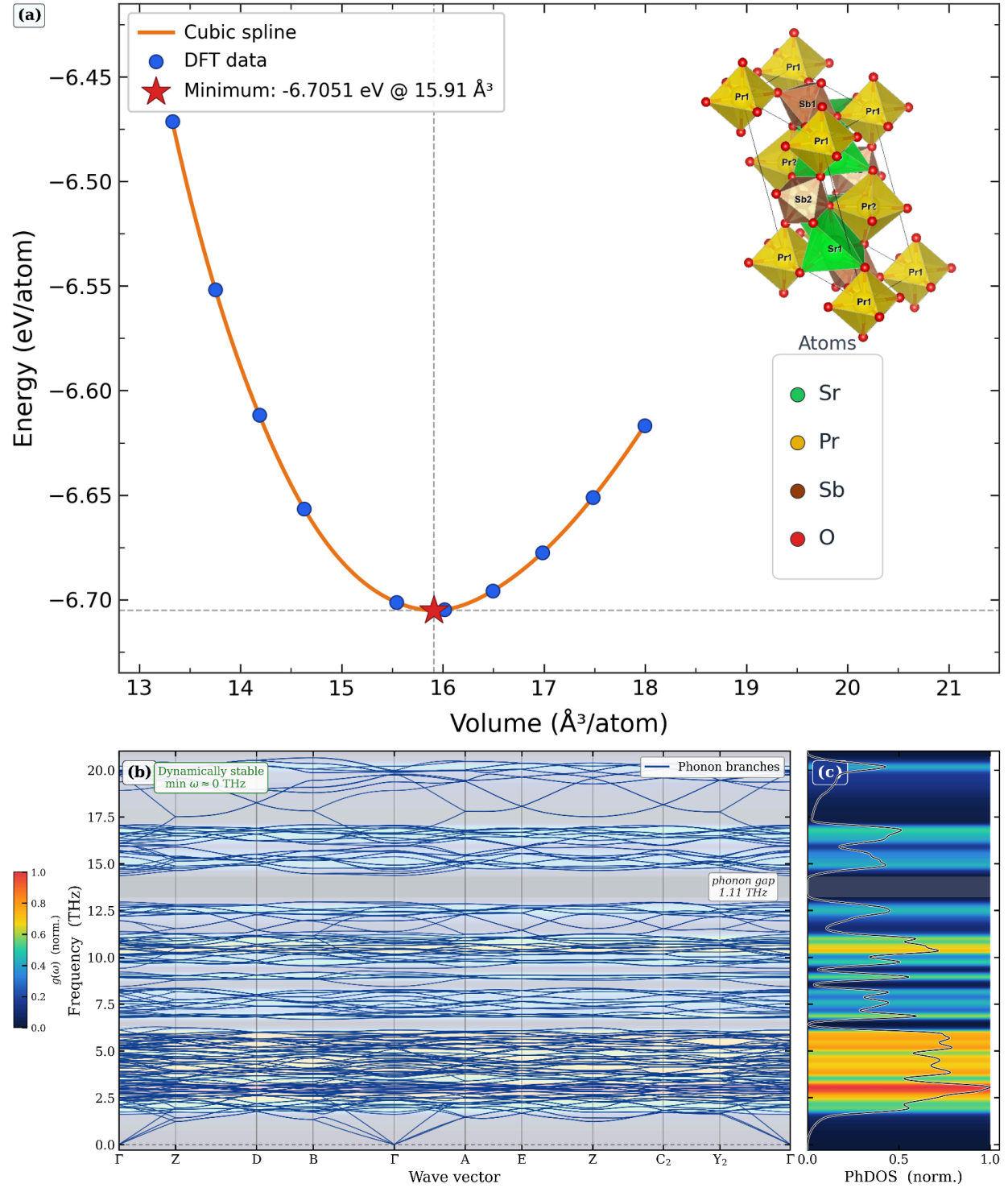


Fig. 1. (a) Energy per atom as a function of volume per atom obtained from DFT calculations (filled circles) fitted with a cubic spline (solid curve). The crystal structure of $\text{Sr}_2\text{PrSbO}_6$ in the double perovskite arrangement is shown

as an inset, with Sr (green), Pr (yellow), Sb (brown), and O (red) atoms labeled. (b) Phonon dispersion of $\text{Sr}_2\text{PrSbO}_6$ along the monoclinic Brillouin zone path $\Gamma \rightarrow Z \rightarrow D \rightarrow B \rightarrow \Gamma \rightarrow A \rightarrow E \rightarrow Z \rightarrow C_2 \rightarrow Y_2 \rightarrow \Gamma$. (c) Atom-projected phonon density of states (PhDOS).

3.2.2. Mechanical Stability

Mechanical stability of the monoclinic $P2_1/c$ structure was verified by computing the complete 6×6 elastic stiffness tensor C_{ij} . For monoclinic symmetry (point group $2/m$), the general stability conditions of Born and Huang⁶⁶ require that all six eigenvalues of C_{ij} be strictly positive. The computed C_{ij} yields eigenvalues of 48.7, 57.3, 79.1, 95.2, 158.6, and 365.0 GPa, confirming the mechanical stability of the structure.

3.2.3. Dynamic Stability

The phonon dispersion along the $\Gamma \rightarrow Z \rightarrow D \rightarrow B \rightarrow \Gamma \rightarrow A \rightarrow E \rightarrow Z \rightarrow C_2 \rightarrow Y_2 \rightarrow \Gamma$ path in Fig. 1(b) exhibits no imaginary frequencies anywhere in the Brillouin zone. This confirms the dynamic stability of the $P2_1/c$ phase. This demonstrates that the relaxed structure corresponds to a true local minimum on the Born–Oppenheimer surface. The phonon spectrum spans from 0 to approximately 21 THz: acoustic modes dominated by the heavy Sr and Pr cations occupy the low-frequency region (0–5 THz). On the other hand, high-frequency optical branches above 15 THz are associated with O-stretching modes within the PrO_6 and SbO_6 octahedra.

3.3. Electronic Band Structure and Density of States

The HSE06 electronic band structure of $\text{Sr}_2\text{PrSbO}_6$ is shown in Fig. 2 alongside the total and projected density of states. The compound is an indirect band gap insulator with a gap $E_g = 5.090$ eV between the valence band maximum (VBM) at the D point ($0, \frac{1}{2}, \frac{1}{2}$) and the conduction band minimum (CBM) at the Γ point ($0, 0, 0$). The 5.09 eV gap places $\text{Sr}_2\text{PrSbO}_6$ in the deep-ultraviolet-transparent class of wide-gap oxides.

The projected density of states in Fig. 2(c) clarifies the orbital character of the band edges. The valence band is dominated by O-2p states ($\approx 87\%$ of the projected weight), extending from approximately -4.3 eV to the VBM at 0 eV, below which an isolated Sb-5s–O bonding manifold lies near -7 eV. The conduction-band minimum at Γ is an antibonding state of mixed Sb-5s ($\approx 35\%$ of the projected weight of the CBM state), Pr-5d ($\approx 29\%$) and O-2p ($\approx 24\%$) character, so that the fundamental gap corresponds to an O-2p $\rightarrow$ Sb-5s/Pr-5d charge-transfer excitation. Higher in the conduction band the density of states is governed first by the comparatively flat Pr-5d manifold, which produces the pronounced peak ≈ 1.9 eV above the CBM, and then by Sr-4d states above ≈ 8 eV. With the Pr_3 potential the $4f^2$ shell of Pr^{3+} is treated as frozen core, consistent with its non-bonding, strongly localised character in the trivalent state; the 4f states therefore do not enter the present description of the band edges.

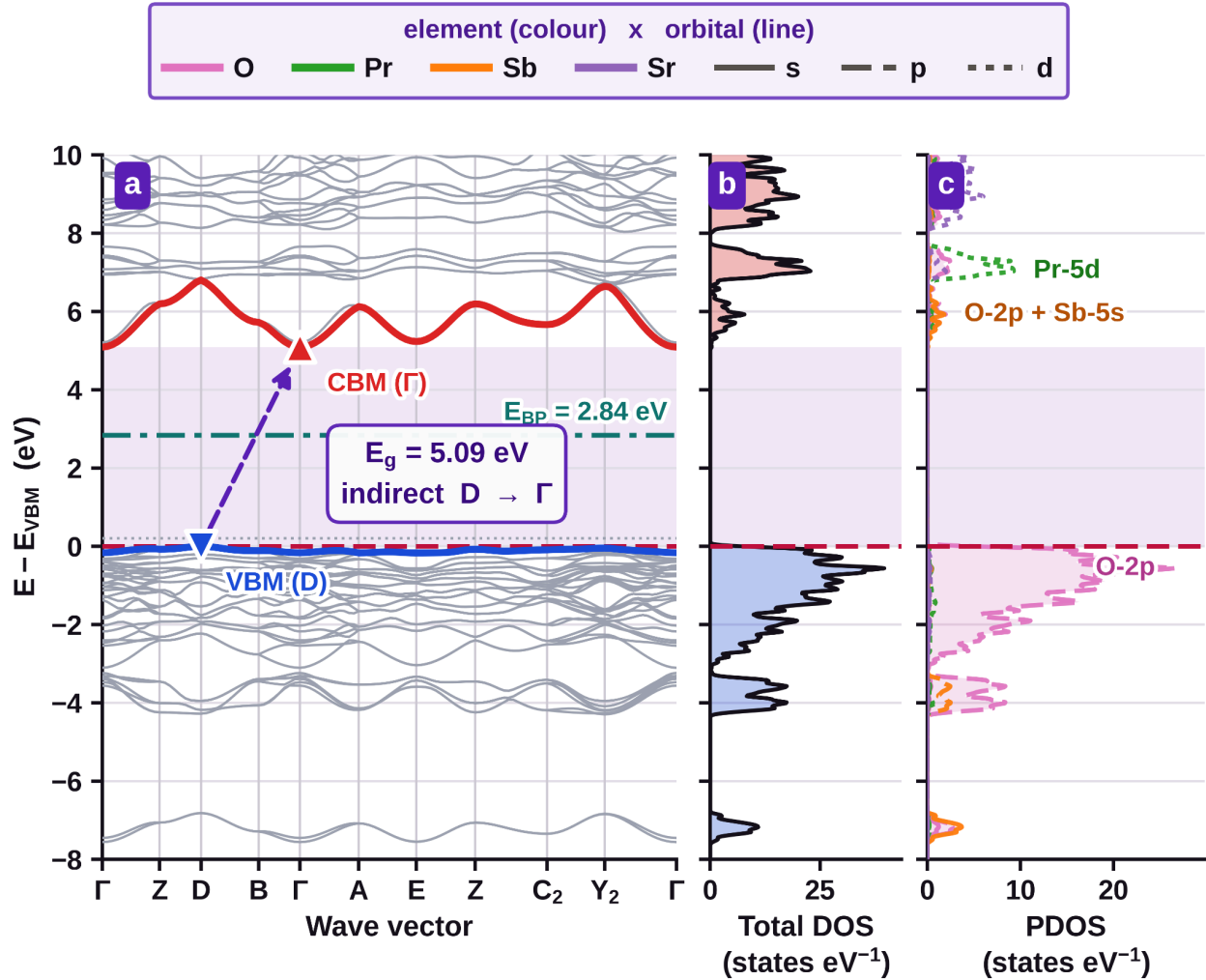


Fig. 2. HSE06 electronic structure of $\text{Sr}_2\text{PrSbO}_6$. (a) Band structure along the monoclinic Brillouin-zone path Γ -Z-D-B- Γ -A-E-Z- C_2 - Y_2 - Γ , with the highest valence and lowest conduction bands highlighted. The energy zero is the valence band maximum (VBM), marked by the dashed horizontal line in all three panels, and the shaded band spans the gap. The compound is an indirect-gap insulator, $E_g = 5.090$ eV, between the VBM at D and the conduction band minimum (CBM) at Γ (dashed arrow). In (a) the faint dotted line is the self-consistent Fermi level and the dash-dot line the branch-point energy, $E_{BP} = 2.84$ eV above the VBM (Section 3.3.1, Table S16). (b) Total electronic density of states (DOS), shaded below the VBM and above the CBM. (c) Orbital-projected DOS (PDOS); the line colour denotes the element and the line style the orbital (solid s, dashed p, dotted d), and the dominant character of each feature is labelled: O-2p states dominate the upper valence band, the conduction-band edge is of mixed Sb-5s, Pr-5d and O-2p character, and the pronounced peak ≈ 1.9 eV above the CBM is Pr-5d derived, with Sr-4d states appearing above ≈ 8 eV.

3.3.1 Branch-point energy and intrinsic dopability

Beyond the magnitude of the gap, the position of the branch-point energy E_{BP} indicates whether a semiconductor can be doped n-type, p-type, or neither.^{46,47} From the HSE06 band structure, E_{BP} lies 2.84 eV above the VBM and 2.25 eV below the CBM, i.e. at 0.56 of the gap (Fig. 2, dotted line; Table S16). An E_{BP} located deep within the gap, rather than near or above the CBM as in readily n-dopable oxides such as ZnO or In_2O_3 ,⁴⁷ places $\text{Sr}_2\text{PrSbO}_6$ in the class of intrinsically insulating ultrawide-gap oxides: extrinsic dopants are expected to produce deep rather than

shallow levels for both carrier types, with only a mild preference for n-type behaviour (E_{BP} is 0.29 eV above mid-gap). The band-edge effective masses reinforce this asymmetry. The CBM is dispersive (bandwidth 1.71 eV; $m_e^* = 0.53\text{--}0.64 m_0$ along $\Gamma\text{--}Z$, $\Gamma\text{--}B$, $\Gamma\text{--}A$ and $\Gamma\text{--}Y_2$), whereas the O-2p-derived valence-band maximum is almost flat (total bandwidth 0.17 eV; $m_h^* \approx 2.6\text{--}2.8 m_0$ along $D\text{--}Z$ and $D\text{--}B$). Such a narrow, heavy O-2p valence band is the electronic signature of hole self-trapping, the mechanism that prevents p-type conduction in $\beta\text{-Ga}_2\text{O}_3$ and related oxides,⁶⁷ and p-type doping of $\text{Sr}_2\text{PrSbO}_6$ is therefore not anticipated. Aligning the bands with the electron affinity ($\chi = 3.5$ eV [see Table S9]) gives the same conclusion on the absolute scale: the CBM lies ≈ 1.4 eV above and the VBM ≈ 3.7 eV below the universal Fermi-stabilisation level (≈ 4.9 eV below vacuum),⁶⁸ so both doping directions are strongly limited by compensating native defects, n-type less severely than p-type.

3.4. Optical Properties

The optical properties of $\text{Sr}_2\text{PrSbO}_6$ are presented in Fig. 3. The real part of the dielectric function $\epsilon_1(\omega)$ shows the xx, yy, and zz components are nearly isotropic throughout the visible and near-UV range, reaching a static value $\epsilon_1(0) \approx 3.5$. The imaginary part $\epsilon_2(\omega)$ exhibits zero absorption for wavelengths exceeding below 243.6 nm, confirming the solar-blind UV-C character of the material. The absorption coefficient α rises steeply for photon energies above the gap, reflecting the direct-allowed transitions that lie ~ 0.5 eV above the indirect fundamental gap. The refractive index $n(\omega) \approx 2.0$ across the visible range indicates moderate optical density, and the reflectivity $R(\omega) \approx 5\%$ at normal incidence is well-suited for transparent applications.

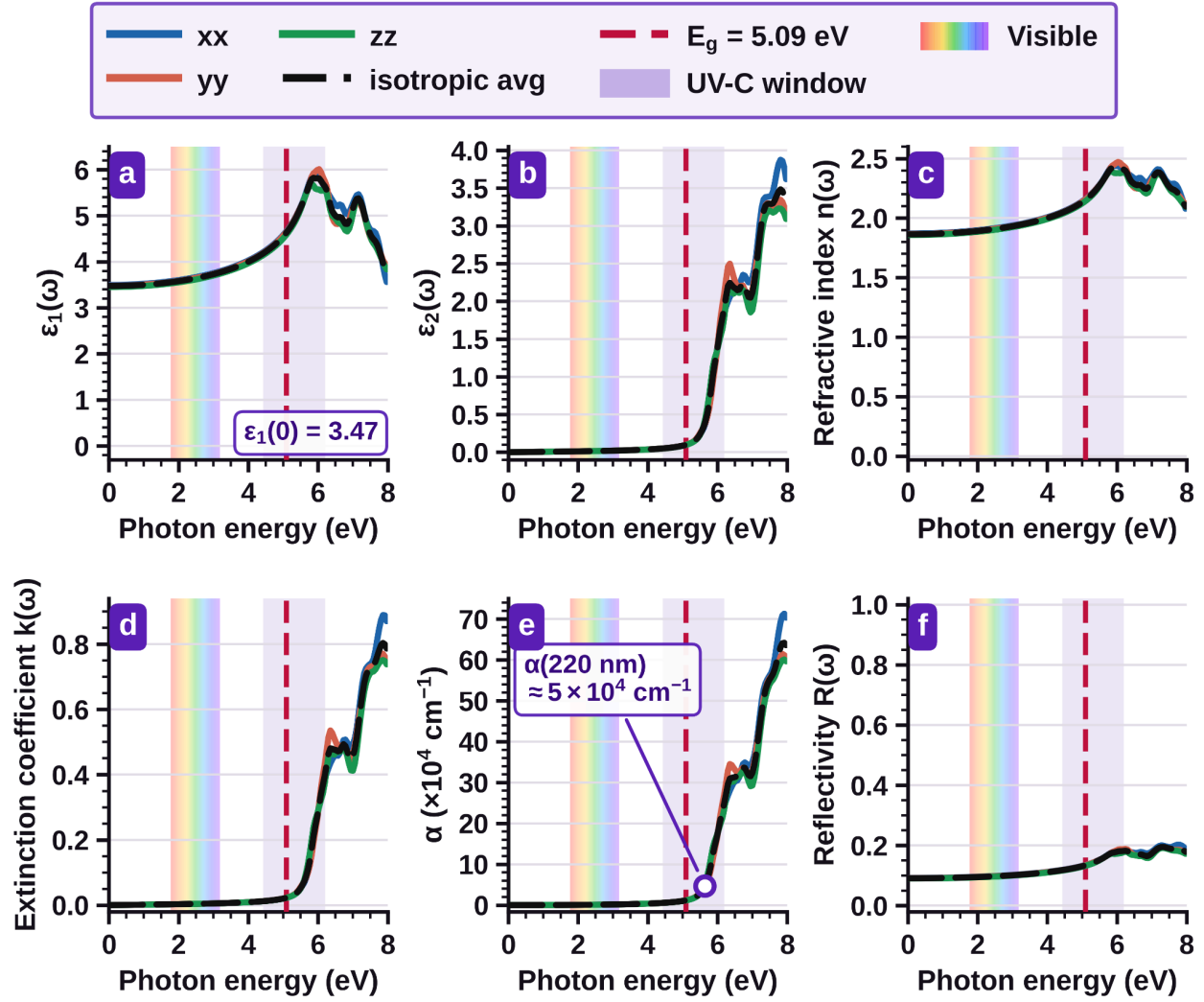


Fig. 3. Optical properties of $\text{Sr}_2\text{PrSbO}_6$. (a) Real part $\epsilon_1(\omega)$ of the dielectric function; (b) imaginary part $\epsilon_2(\omega)$; (c) refractive index $n(\omega)$; (d) extinction coefficient $k(\omega)$; (e) absorption coefficient; (f) normal-incidence reflectivity $R(\omega)$. All three polarisation components (xx, yy, zz) and the isotropic average (avg) are shown. Visible range is indicated by the rainbow shading (390–700 nm) and the solar-blind UV-C window by the violet shading (200–280 nm); the dashed vertical line marks the HSE06 indirect gap ($E_g = 5.09$ eV), and the open circle in (e) marks the 220 nm operating point used in the device simulations ($\alpha \approx 5 \times 10^4 \text{ cm}^{-1}$).

3.5. Mechanical Properties

The full 6×6 elastic stiffness tensor for the monoclinic $\text{P2}_1/\text{c}$ structure contains 13 independent components C_{ij} . The diagonal elements $C_{11} = 191.8$ GPa, $C_{22} = 202.8$ GPa, and $C_{33} = 221.7$ GPa indicate strong resistance to uniaxial compression in all three crystallographic directions. The off-diagonal stiffness constants $C_{12} = 102.4$ GPa and $C_{13} = 70.1$ GPa indicate significant cross-directional coupling. The shear moduli $C_{44} = 60.5$ GPa, $C_{55} = 51.1$ GPa, and $C_{66} = 75.9$ GPa confirm resistance to shear deformation.

The polycrystalline elastic moduli from the Voigt–Reuss–Hill (VRH) averaging scheme are summarised in Table 2. The Hill averages are: bulk modulus $B_H = 121.6$ GPa, shear modulus $G_H = 60.9$ GPa, Young's modulus $E_H = 156.5$ GPa, and Poisson's ratio $\nu_H = 0.285$. The Pugh ratio $B/G = 2.00$ exceeds the empirical ductile/brittle threshold of 1.75^{69} , classifying $\text{Sr}_2\text{PrSbO}_6$ as a

ductile oxide. The Vickers hardness estimated using the Chen⁷⁰ and Tian⁷¹ empirical relations yields $H_V = 6.8$ and 7.7 GPa, respectively. The Cauchy pressure $C_{12} - C_{44} = 41.9$ GPa is positive, indicating mixed ionic–covalent bonding character⁷². The Debye temperature derived from the elastic wave velocities ($v_L = 5915$ m/s, $v_T = 3241$ m/s) is $\theta_D = 433.4$ K.

Table 2. Calculated polycrystalline elastic moduli and related quantities for $\text{Sr}_2\text{PrSbO}_6$ from the Voigt–Reuss–Hill (VRH) averaging scheme (GGA-PBE).

Property	Voigt	Reuss	Hill
Bulk modulus B (GPa)	121.61	121.51	121.56
Shear modulus G (GPa)	62.65	59.09	60.87
Young's modulus E (GPa)	160.41	152.54	156.49
Poisson's ratio ν	0.280	0.291	0.285
P-wave modulus M (GPa)	205.14	200.30	202.72
Pugh's ratio B/G	1.94	2.06	2.00
Vickers hardness H_V (Chen) (GPa)	7.30	6.31	6.80
Vickers hardness H_V (Tian) (GPa)	8.10	7.28	7.69
Cauchy pressure P_c (GPa)	—	—	41.9
Univ. anisotropy index A^U	—	—	0.30
Debye temperature θ_D (K)	—	—	433.4
Longitudinal velocity v_L (m/s)	—	—	5915
Transverse velocity v_T (m/s)	—	—	3241

The thermoelectric transport properties of $\text{Sr}_2\text{PrSbO}_6$, computed by solving the phonon Boltzmann transport equation (phono3py) together with BoltzTraP electronic transport calculations, are presented in Supplementary Note 1 (Figs. S11–S14). Briefly, the lattice thermal conductivity is high and anisotropic ($\kappa_{\text{avg}} = 41.7$ W m^{−1} K^{−1} at 300 K, decreasing to 22.3 W m^{−1} K^{−1} at 1100 K) and is carried almost entirely (99.6%) by optical phonon branches, while p-type transport yields a modest maximum figure of merit $ZT \approx 0.14$ at 1100 K. In the photodetector context, the high thermal conductivity favours efficient heat dissipation under UV irradiation, and the appreciable Seebeck coefficient (185–200 $\mu\text{V K}^{-1}$) may be exploited for thermally modulated or self-powered sensing schemes.

3.6 Predicted Solar-Blind UV Photodetector Performance

The solar-blind ultraviolet (UV) spectral window, defined as the wavelength range 200–280 nm, is completely absorbed by atmospheric ozone and molecular oxygen and therefore exhibits zero solar background at ground level. Photodetectors operating exclusively within this window offer

inherent noise immunity against solar radiation, enabling high-contrast detection for applications including missile plume identification, flame sensing, space-to-ground optical communications, corona discharge monitoring, and environmental surveillance^{4,5}. Wide bandgap semiconductors with band gaps exceeding ~ 4.43 eV (corresponding to $\lambda < 280$ nm) are candidates for solar-blind detection, and materials with gaps exceeding ~ 5.0 eV provide an even narrower, deeper UV-C (100 nm $\sim$ 280 nm) response window with enhanced selectivity.

3.6.1 Device Architecture and Band Diagram

The simulated device consists of a planar MSM photoconductor in which two platinum (Pt) Schottky contacts are deposited on the surface of an $\text{Sr}_2\text{PrSbO}_6$ thin film, separated by an interelectrode gap of $L = 10\ \mu\text{m}$ [Fig.4 (a)]. The MSM geometry is the most widely adopted architecture for wide-bandgap UV photodetectors owing to its fabrication simplicity, inherently low dark current (arising from back-to-back Schottky junctions), and compatibility with thin-film deposition on arbitrary substrates^{51,73}.

Platinum was selected as the contact metal for its high work function ($\Phi_M = 5.65$ eV).^[9] Fig. 4(b) shows the equilibrium ($V = 0$) band alignment. Because the film is undoped, its intrinsic carrier density ($n_i \approx 3 \times 10^{-24}\ \text{cm}^{-3}$ for $E_g = 5.09$ eV) is far too small to screen any band bending: the Debye length, $L_D = (\epsilon k_B T / q^2 n_i)^{1/2} \approx 10^{12}$ m, and even the depletion width for a hypothetical unintentional donor density of $10^{12}\ \text{cm}^{-3}$ ($\approx 12\ \mu\text{m}$) exceed the $10\ \mu\text{m}$ electrode gap. The bands are therefore flat across the entire film and the Fermi level of the whole device coincides with that of the Pt electrodes, $E_F = -\Phi_M = -5.65$ eV, i.e. 0.4 eV above mid-gap. With the electron affinity $\chi = 3.5$ eV this places E_F at $\Phi_{B,n} = \Phi_M - \chi = 2.15$ eV below the conduction-band minimum and $\Phi_{B,p} = E_g - \Phi_{B,n} = 2.94$ eV above the valence-band maximum throughout the film, corresponding to equilibrium carrier densities $n \approx 7 \times 10^{-18}\ \text{cm}^{-3}$ and $p \approx 10^{-30}\ \text{cm}^{-3}$. The Schottky–Mott value $\Phi_{B,n} = 2.15$ eV assumes no Fermi-level pinning. In the opposite (Bardeen) limit the interface Fermi level is pinned at the branch-point energy, giving $\Phi_{B,n} = E_C - E_{BP} = 2.25$ eV (Section 3.3.1); with the empirical pinning factor $S = [1 + 0.1(\epsilon_\infty - 1)^2]^{-1} \approx 0.53$ for $\epsilon_\infty = n^2 \approx 4$, the interpolated barrier is 2.20 eV. The barrier height is thus 2.15–2.25 eV irrespective of the degree of pinning, and the thermionic dark current over it ($\approx 10^{-35}$ A, Section 2.6) is negligible at 300 K, consistent with the ultralow dark currents of wide-band-gap MSM devices.⁷⁴ Under bias the uniform field V/L tilts the bands linearly across the gap [Fig. 4(c)], and under 220 nm illumination the photogenerated population $n \approx p \approx 10^{10}\ \text{cm}^{-3}$ splits the quasi-Fermi levels to $E_C - E_{Fn} \approx 0.49$ eV and $E_{Fp} - E_V \approx 0.51$ eV.

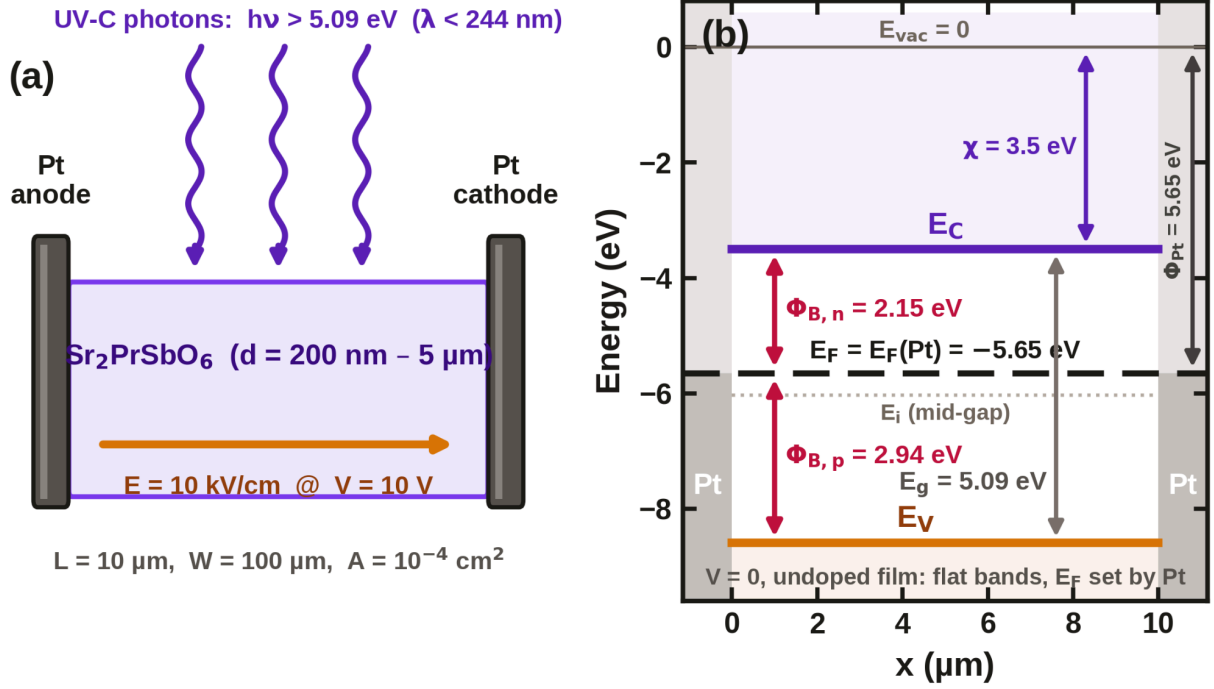


Fig. 4: (a) Schematic of the Pt/Sr₂PrSbO₆/Pt MSM photoconductor under UV-C illumination and applied field. (b) Equilibrium ($V = 0$) band diagram. For the undoped film the bands are flat across the 10 μm gap and a single Fermi level, $E_F = E_F(Pt) = -5.65$ eV, is shared by the electrodes and the film, lying $\Phi_{B,n} = 2.15$ eV below E_C and $\Phi_{B,p} = 2.94$ eV above E_V ($\chi = 3.5$ eV, $E_g = 5.09$ eV, HSE06); the intrinsic level E_i is shown for reference. (c) Band diagram at $V = 10$ V under 220 nm illumination (1 mW cm⁻²): the bands tilt uniformly ($|E| = V/L = 10$ kV cm⁻¹) and the quasi-Fermi levels E_{Fn} , E_{Fp} split by the photogenerated carrier density $n \approx p \approx 10^{10}$ cm⁻³ from the DEVSIM simulation.

3.6.2 I–V Characteristics and Spatial Profiles

Fig. 5 shows the current–voltage (I–V) characteristics of the Sr₂PrSbO₆ MSM device under dark and UV-illuminated conditions, computed by the DEVSIM TCAD solver. The dark current remains below 10^{-24} A over the full bias range of 0–20 V, reflecting the extreme suppression of carrier transport by the large Schottky barrier ($\Phi_B = 2.15$ eV) and the essentially zero intrinsic carrier density arising from the 5.09 eV band gap.

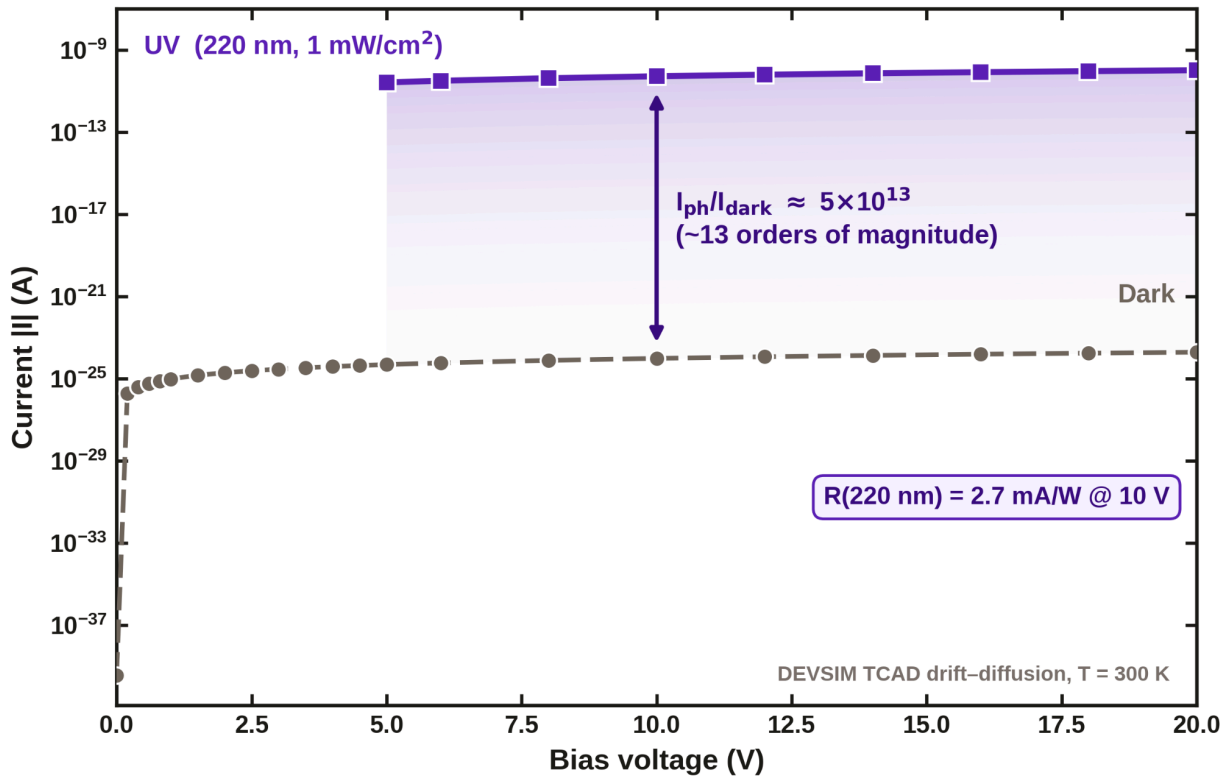


Fig. 5. I–V Dark and UV-illuminated I–V characteristics from DEVSIM TCAD drift-diffusion simulation at $T = 300$ K. The photocurrent exceeds the intrinsic dark current by ~ 13 orders of magnitude, corresponding to $R(220 \text{ nm}) = 2.7 \text{ mA/W}$ for the 200 nm baseline device.

Fig. 6(a–c) presents the spatial distributions of the electric field, carrier concentrations, and optical generation rate obtained from the simulation at $V = 10 \text{ V}$. In the dark state, the electric field is uniform at $|E| = V/L = 10 \text{ kV/cm}$ throughout the device interior, reflecting the charge-neutral nature of the near-intrinsic semiconductor. Under 220 nm illumination, the photogenerated carrier density reaches $n \approx p \approx 10^{10} \text{ cm}^{-3}$ in the device bulk. Carrier densities fall to their equilibrium values within the last mesh cells at the two contacts, where the boundary condition fixes n and p ; this is carrier extraction at the electrodes, not a depletion region — the undoped film cannot sustain band bending (Section 3.5.1). The optical generation profile [Fig. 6(c)] calculated from the Beer–Lambert law shows a gradual decline with increasing depth.

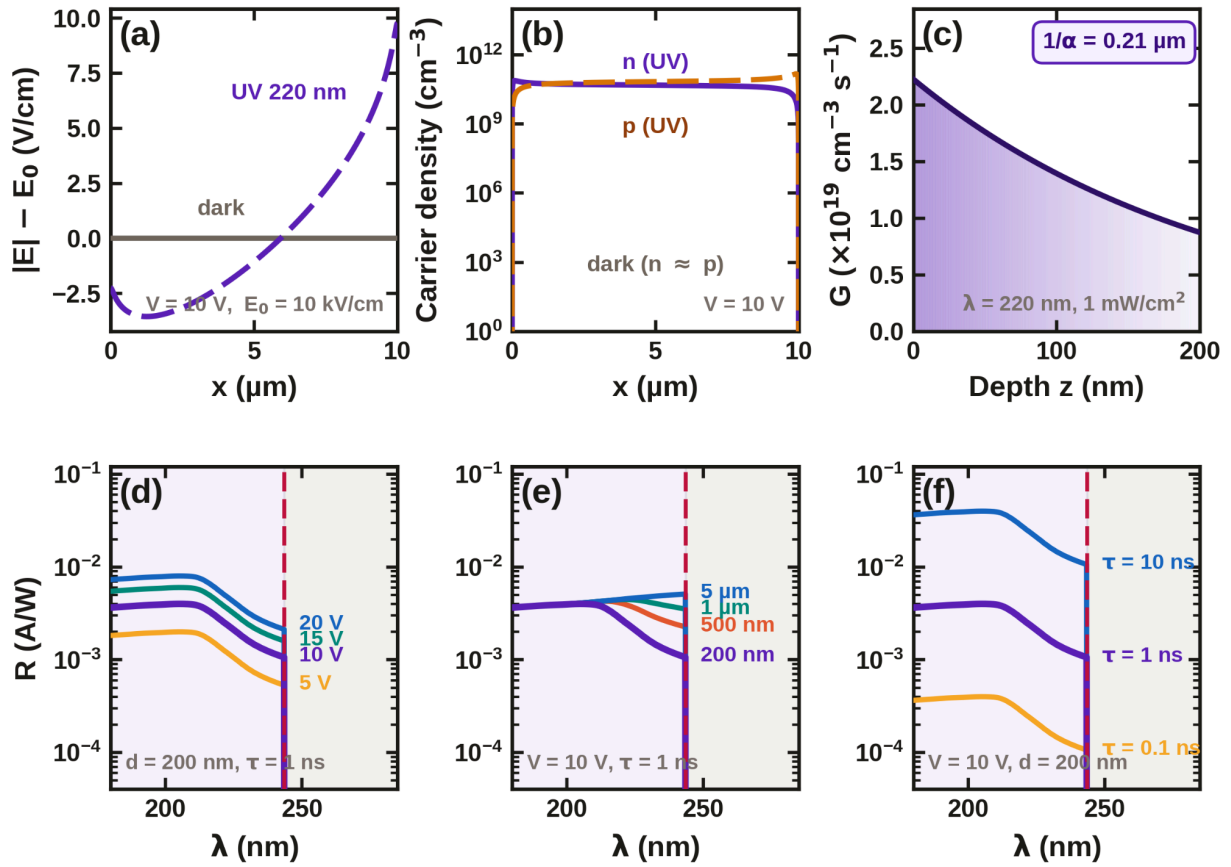


Fig. 6. (a) Electric-field deviation $|E| - E_0$ across the electrode gap in the dark and under 220 nm illumination at $V = 10$ V. (b) Electron and hole density profiles in the dark and under illumination. (c) Beer-Lambert optical generation profile through the film depth at 220 nm. (d-f) Spectral responsivity $R(\lambda)$ as a function of (d) bias voltage (5–20 V), (e) film thickness (200 nm – 5 μm), and (f) carrier lifetime (0.1–10 ns); the thick curve marks the baseline device ($V = 10$ V, $d = 200$ nm, $\tau = 1$ ns). Responsivity vanishes above $\lambda = 244$ nm, confirming intrinsic solar-blind operation.

3.6.3 Spectral Responsivity and Solar-Blind Performance

The spectral responsivity $R(\lambda) = \eta_{\text{abs}} \times G_{\text{pc}} / h\nu$ was computed over the full UV–visible range (180–600 nm), where $G_{\text{pc}} = (\mu_e + \mu_h)\tau V/L^2$ is the photoconductor gain. Figure 10 shows $R(\lambda)$ as functions of bias voltage (panel d), film thickness (panel e), and carrier lifetime (panel f). At $V = 10$ V with a 200 nm film and $\tau = 1$ ns, $R(220 \text{ nm}) = 0.061 \text{ mA/W}$. The responsivity is strictly zero for all $\lambda > 243.6$ nm, confirming theoretically perfect solar-blind rejection that is a direct consequence of the HSE06-computed band gap.

The HSE06 optical calculation reveals an absorption coefficient at the primary operating wavelength of $\alpha(220 \text{ nm}) = 5.02 \times 10^4 \text{ cm}^{-1}$ arising from direct-allowed optical transitions at higher symmetry points located 0.55 eV above the indirect fundamental gap. This yields a film absorption efficiency $\eta_{\text{abs}} \approx 53\%$ for the 200 nm baseline device, removing absorption as the dominant performance-limiting mechanism. The principal constraint on responsivity at baseline conditions is instead the photoconductive gain, $G_{\text{pc}} = 0.03$, which reflects a short carrier lifetime ($\tau = 1$ ns) relative to the carrier transit time ($t_{\text{tr}} = L^2/\mu_n V \approx 50$ ns), so that most photogenerated

carriers recombine before collection. Increasing the film thickness to $d = 5 \mu\text{m}$ raises η_{abs} to $\sim 84\%$ as the film approaches optical saturation ($\alpha d \gg 1$) and improves R to 4.5 mA/W [Fig. 6(e)], with diminishing returns beyond this thickness. Further optimization through bias enhancement ($V \rightarrow 20 \text{ V}$) extends R to 8.9 mA/W , while longer lifetimes ($\tau = 10 \text{ ns}$, [Fig. 6(f)]) yield $R \approx 44 \text{ mA/W}$ at $d = 5 \mu\text{m}$, substantially exceeding the performance of state-of-the-art Ga_2O_3 MSM devices.^[14]

3.6.4 External Quantum Efficiency and Specific Detectivity

The external quantum efficiency $\text{EQE}(\lambda) = R(\lambda) \times h\nu/q \times 100\%$ reaches 1.53% at 220 nm ($V = 10 \text{ V}$, $d = 200 \text{ nm}$). With $\eta_{\text{abs}} \approx 53\%$ at 200 nm , optical absorption is not the limiting mechanism rather the remaining efficiency gap is governed entirely by the photoconductive gain $G_{\text{pc}} = 0.03$, which arises because the carrier lifetime ($\tau = 1 \text{ ns}$) is short relative to the transit time ($t_{\text{tr}} \approx 50 \text{ ns}$).

The specific detectivity $D^* = R\sqrt{A} / \sqrt{(2qJ_{\text{dark}})}$ was evaluated using the practical dark current density $J_{\text{dark}} = 10^{-10} \text{ A/cm}^2$ adopted from experimentally reported values for $\beta\text{-Ga}_2\text{O}_3$ MSM Schottky devices.⁵⁰ This yields $D^*(220 \text{ nm}) = 1.6 \times 10^9 \text{ Jones}$ for the 200 nm baseline device, improving to $\sim 2.5 \times 10^9 \text{ Jones}$ at $d = 5 \mu\text{m}$ as η_{abs} saturates [Fig. 7(b)].

For reference, Fig. 7(b) also shows the theoretical upper bound D^* computed using the thermionic emission dark current (ideal Schottky junction, no defect-assisted leakage), which gives $D^* \approx 10^{17} \text{ Jones}$. The $\sim 10^8$ -order gap between this ideal limit and the practical estimate underscores the importance of using a literature-calibrated J_{dark} when reporting D^* for comparison with experimental devices.

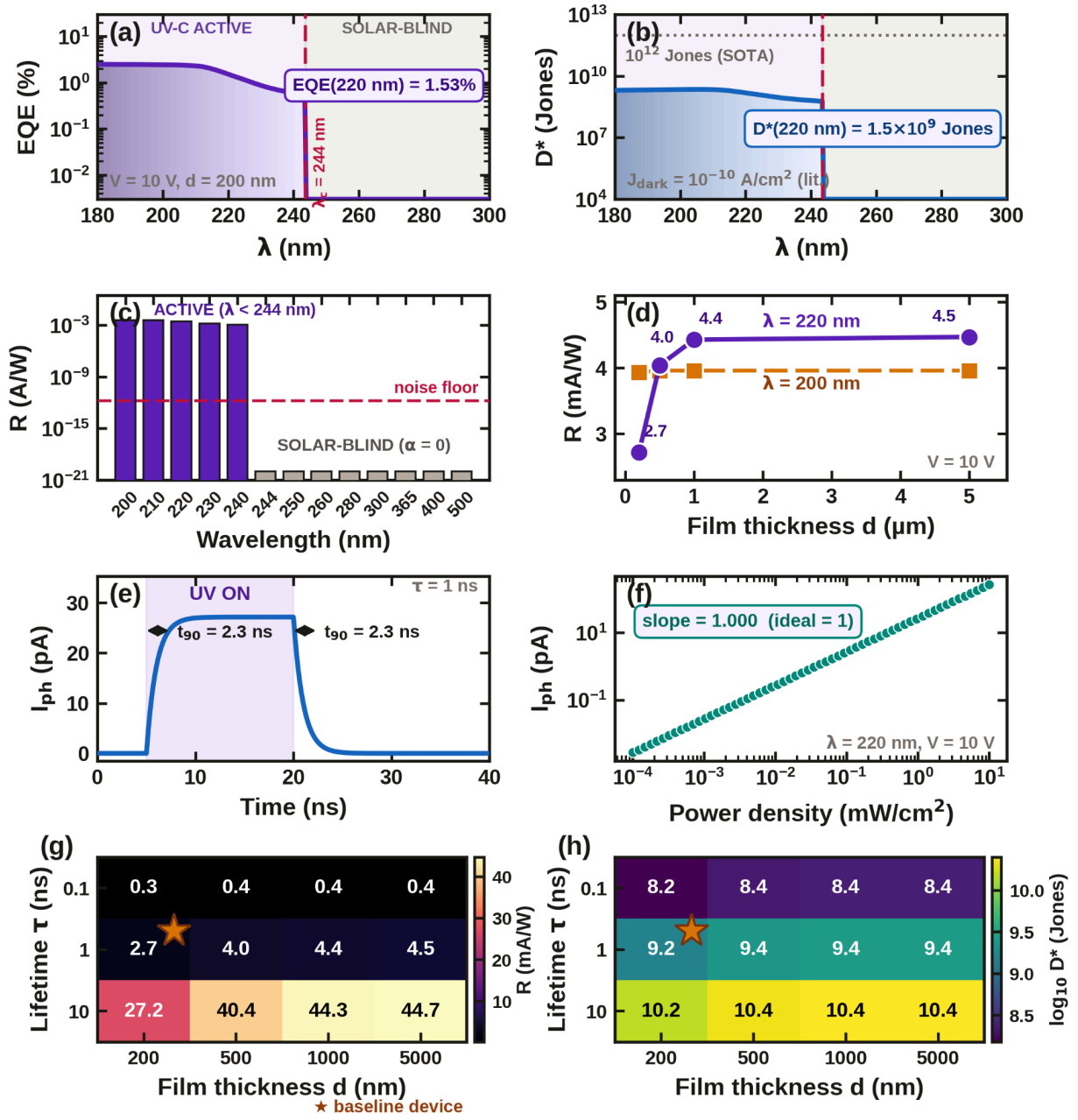


Fig. 7: (a) External quantum efficiency, $\text{EQE}(220 \text{ nm}) = 1.53\%$. (b) Specific detectivity $D^*(220 \text{ nm}) = 1.5 \times 10^9$ Jones. (c) Responsivity at key wavelengths: finite response for $\lambda < 244 \text{ nm}$ and zero response beyond the cutoff, demonstrating full solar-blind rejection. (d) Responsivity versus film thickness at 220 nm and 200 nm. (e) Transient photoresponse with $\tau = 1 \text{ ns}$, giving t_{90} rise and fall times of 2.3 ns. (f) Photocurrent linearity over five decades of incident power (slope = 1.000). (g, h) Sensitivity maps of responsivity and $\log_{10}(D^*)$ versus film thickness and carrier lifetime; the star marks the baseline device.

3.6.5 Solar-Blind Rejection Performance

Fig. 7(c) displays the responsivity at key wavelengths spanning the solar-blind UV-C ($\lambda \leq 244$ nm), UV-B (280–315 nm), UV-A (315–400 nm), and visible (400–500 nm) regions. All wavelengths above 243.6 nm exhibit zero responsivity, while UV-C wavelengths between 200 and 243 nm show measurable response, with $R(220 \text{ nm}) = 2.82 \text{ mA/W}$ at $V = 10 \text{ V}$ and $d = 200 \text{ nm}$. This demonstrates a theoretically infinite UV/visible rejection ratio from first principles, arising from the exact vanishing of the absorption coefficient at the HSE06 band gap energy.

Fig. 7(d) shows that responsivity increases steeply with film thickness at small d but saturates rapidly beyond $d \approx 1 \mu\text{m}$. This behaviour reflects the large absorption coefficient $\alpha(220 \text{ nm}) = 5.02 \times 10^4 \text{ cm}^{-1}$ as η_{abs} reaches $\sim 53\%$ at $d = 200 \text{ nm}$, $\sim 83\%$ at $d = 1 \mu\text{m}$, and $\sim 84\%$ at $d = 5 \mu\text{m}$, with diminishing returns thereafter. Although $\text{Sr}_2\text{PrSbO}_6$ possesses an indirect fundamental gap at 5.09 eV, operation at 220 nm accesses direct-allowed optical transitions at higher symmetry points located 0.55 eV above the gap onset, producing absorption coefficients comparable to those of direct-gap materials such as $\beta\text{-Ga}_2\text{O}_3$ or AlGaN. The spectral selectivity of $\text{Sr}_2\text{PrSbO}_6$ remains superior to these counterparts owing to its wider band gap and the resulting hard solar-blind cutoff at 243.6 nm.^{8,75,76}

3.8.6 Transient Response and Photocurrent Linearity

The temporal photocurrent response to a pulsed 220 nm UV-C excitation (on at $t = 5 \text{ ns}$, off at $t = 20 \text{ ns}$) is shown in Fig. 7(e). The rise and fall times are both governed by the SRH carrier lifetime, giving $\tau_{\text{rise}} = \tau_{\text{fall}} = \tau = 1 \text{ ns}$ and a 90% response time $t_{90\%} \approx 2.3 \text{ ns}$. The high HSE06 optical generation rate produces a peak photocurrent of $\sim 28 \text{ pA}$ at 1 mW/cm^2 , while the nanosecond timing remains determined solely by τ and is independent of the optical absorption. This nanosecond-scale response is suitable for high-speed UV-C monitoring applications. Longer lifetime devices ($\tau = 10 \text{ ns}$) would trade speed for responsivity.

The log-log photocurrent–power relationship at 220 nm [Fig. 7(f)] is linear over five orders of magnitude in incident power ($0.1 \mu\text{W/cm}^2$ to 10 mW/cm^2), with a fitted slope of 1.0000, confirming strictly ohmic photoconductor behavior with no saturation or recombination nonlinearity in this power range.

3.6.7 Sensitivity Analysis

Fig. 7(g,h) summarises the responsivity and detectivity as functions of both film thickness and carrier lifetime in a heat-map format, providing a design roadmap for experimental realisation. Responsivity ranges from 0.28 mA/W ($\tau = 0.1 \text{ ns}$, $d = 200 \text{ nm}$) to 44.5 mA/W ($\tau = 10 \text{ ns}$, $d = 5 \mu\text{m}$). The corresponding $\log_{10}(D^*)$ spans 8.2 to 10.4, indicating that D^* in the range $10^{8.2} - 10^{10.4}$ Jones is achievable depending on material quality and device geometry.

A key feature of the heatmap is that variation along the thickness axis is weak. With $\alpha(220 \text{ nm}) = 5.02 \times 10^4 \text{ cm}^{-1}$, the film absorption efficiency saturates beyond $d \approx 1 \mu\text{m}$, leaving carrier lifetime τ as the overwhelmingly dominant design variable. Improving τ from 0.1 ns to 10 ns yields a $100\times$ gain in both R and D^* at any thickness, whereas increasing d from 200 nm to $5 \mu\text{m}$ at fixed τ produces only a $\sim 1.6\times$ improvement. This shifts the experimental priority from growing thicker films to improving material quality and reducing trap-assisted recombination. These projections

assume the practical dark current density $J_{\text{dark}} = 10^{-10}$ A/cm² and are conservative with respect to the theoretical thermionic limit.

3.6.8 Comparison with State-of-the-Art Solar-Blind Detectors

Fig. 8 places the predicted performance of Sr₂PrSbO₆ on the band-gap–responsivity landscape of reported solar-blind UV detectors. Two axes of comparison must be distinguished: responsivity, a device-engineering figure of merit set largely by photoconductive gain, contact quality, and carrier lifetime; and spectral selectivity, an intrinsic material property fixed by the band gap. Sr₂PrSbO₆ is distinguished by the latter, not the former.

On the responsivity axis the predicted values are modest relative to experimental devices, and it is important to compare like with like [Fig. 8]. Devices without internal gain (EQE ≤ 100 %), including the most recent self-powered AlGaIn, AlYN, NiO/Ga₂O₃, NiO/ZnGa₂O₄ and all-diamond detectors reported in 2023–2026^{77–81}, cluster between 0.01 and 0.1 A/W, i.e. within one to two orders of magnitude of the 2.7–45 mA/W predicted here without any gain mechanism. The responsivities of 10²–10¹⁰ A/W reported for β-Ga₂O₃, amorphous Ga₂O₃, MgZnO/InGaO or AlGaIn devices arise from trap-assisted photoconductive gain, two-dimensional-electron-gas phototransistor action or avalanche multiplication, are typically obtained at μW cm⁻² or lower irradiance, and are accompanied by slower response—the gain–bandwidth trade-off emphasised in recent reviews. The separation between the present prediction and these gain-enhanced devices is therefore not a fundamental limitation of the material but a consequence of the conservative model assumptions; carrier-lifetime engineering and gain mechanisms of the kind exploited in those devices are the principal levers for improvement.

The distinguishing strength of Sr₂PrSbO₆ lies on the selectivity axis. Its HSE06 band gap of 5.09 eV exceeds that of β-Ga₂O₃ (≈4.8 eV), shifting the intrinsic absorption cutoff to $\lambda_c = 243.6$ nm, roughly 15 nm deeper into the UV-C than the ~259 nm edge of Ga₂O₃ and placing the full response within the deep solar-blind window with negligible leakage into the UV-B and UV-A bands. Because this cutoff derives from the fundamental gap rather than from external filtering or device geometry, it is intrinsic, abrupt, and reproducible, giving Sr₂PrSbO₆ a spectral-selectivity advantage that is independent of the responsivity considerations above. For applications in which deep-UV-C discrimination and false-alarm immunity against a strictly zero solar background outweigh raw responsivity such as corona-discharge sensing, flame detection, and non-line-of-sight UV communication. This pairing of a hard, deep cutoff with predicted triple stability constitutes the material's central value proposition. Its deep-gap branch-point energy (Section 3.3.1) further implies that the low dark current is an intrinsic material property rather than a consequence of contact engineering.

The one-dimensional DEVSIM profiles resolve the lateral electric field, carrier density, and Beer–Lambert generation rate $G(z)$ along independent cross-sections, which is sufficient for all macroscopic metrics. To confirm the underlying spatial coupling, 2D FEM maps of $G(x,y)$, $n(x,y)$, and $\phi(x,y)$ across the full 10 μm × 200 nm cross-section are provided in the Supporting Information (Fig. S1–S3). The potential and field maps (Fig. S2) show that $|E_x|$ is spatially uniform throughout the device at ≈10 kV/cm with no depth dependence, directly validating the 1D approximation. The mid-gap $G(y)$ profile (Fig. S3) reproduces the analytical Beer–Lambert

decay quantitatively ($1/\alpha = 213$ nm), and the illuminated carrier map yields $I_{ph}/I_d \approx 5.4 \times 10^{13}$, reflecting the orders-of-magnitude contrast between the wide-gap dark carrier density and the photogenerated population at 220 nm. The static 2D maps capture device behaviour at a single operating point (220 nm, 10 V). Movies S4–S8 (Supporting Information) extend this picture dynamically: Movie S4 shows the SRH-limited carrier rise and fall ($\tau \approx 1$ ns); Movie S5 sweeps 180–400 nm and visualises $G(z, \lambda)$ collapsing to zero at the solar-blind cutoff $\lambda_c = 243.6$ nm; and Movies S6–S8 provide a rotating 3D device view, the $R(\lambda, V)$ responsivity surface, solar-blind operation against the AM1.5 spectrum.

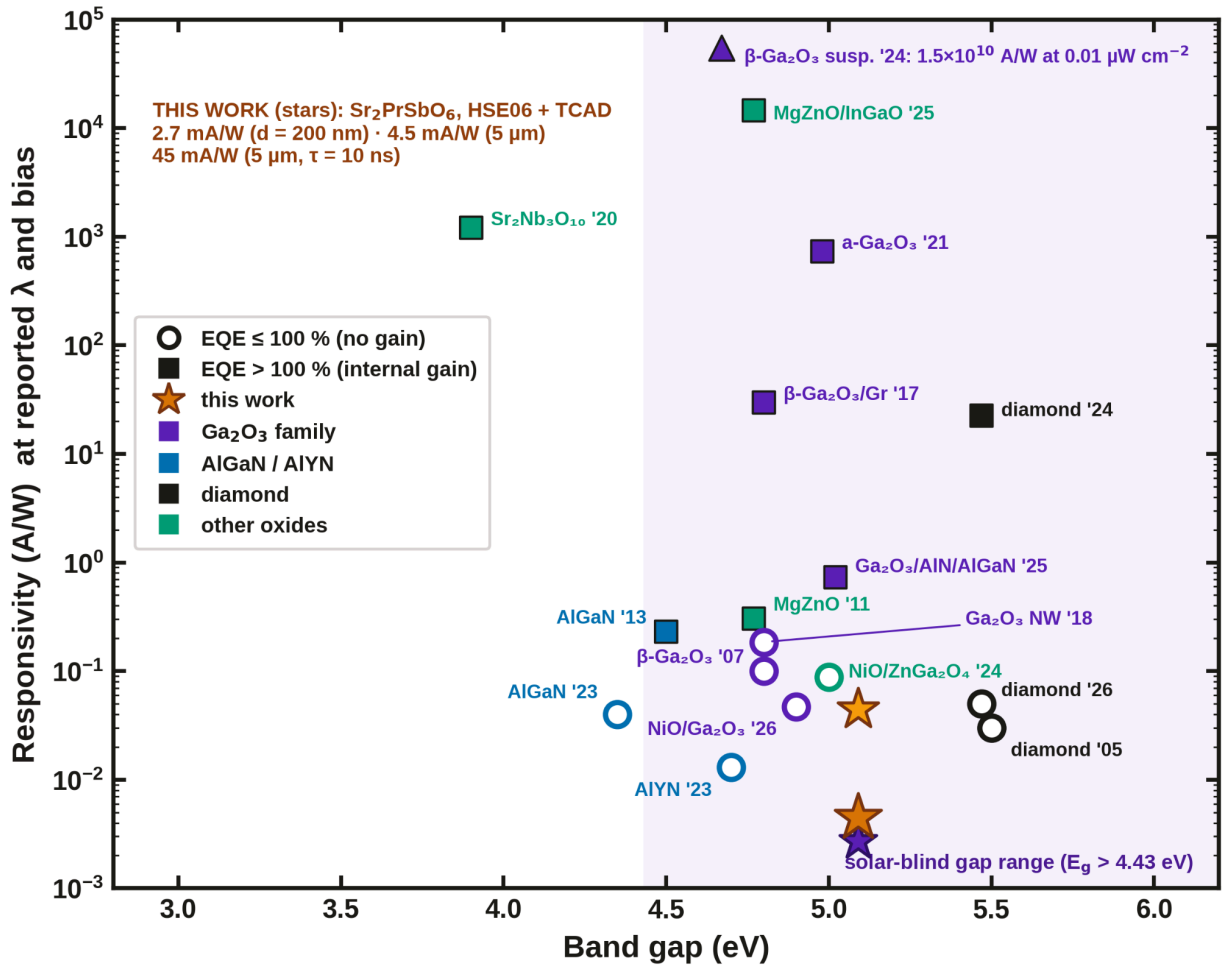


Fig. 8: Benchmark of the predicted $\text{Sr}_2\text{PrSbO}_6$ photodetector against reported solar-blind photodetectors, 2005–2026 (data and sources in Table S19). Responsivity versus band gap, each literature value at its reported wavelength and

bias. Open circles: devices without internal gain ($\text{EQE} \leq 100\%$); filled squares: devices with internal photoconductive, phototransistor or avalanche gain ($\text{EQE} > 100\%$); the suspended-channel $\beta\text{-Ga}_2\text{O}_3$ record (1.5×10^{10} A/W at $0.01 \mu\text{W cm}^{-2}$) lies above the axis range. Stars: this work at 220 nm and 10 V for $d = 200$ nm, $5 \mu\text{m}$, and $5 \mu\text{m}$ with $\tau = 10$ ns. Shading marks the solar-blind band-gap range ($E_g > 4.43$ eV).

4. Conclusions

A comprehensive first-principles investigation of the structural, electronic, mechanical, phonon, thermodynamic, equation-of-state, optical, and thermoelectric transport properties of the monoclinic double perovskite oxide $\text{Sr}_2\text{PrSbO}_6$, together with a first-principles-guided evaluation of its solar-blind UV-C photodetector performance has been presented in this study. The compound adopts the $\text{P2}_1/\text{c}$ (No. 14) space group with lattice parameters $a = 5.887$ Å, $b = 6.004$ Å, $c = 10.214$ Å, $\beta = 124.82^\circ$, and a Goldschmidt tolerance factor $t = 0.915$, consistent with the octahedral-tilt distortion expected for the $\text{Pr}^{3+}/\text{Sb}^{5+}$ 3+/5+ B-site charge combination.

Triple stability is confirmed: the computed formation enthalpy is negative (thermodynamic), all 13 independent monoclinic elastic constants satisfy the Born–Huang criteria, and no imaginary phonon frequencies appear anywhere in the Brillouin zone. The concurrent satisfaction of all three criteria strongly supports the synthesizability of $\text{Sr}_2\text{PrSbO}_6$. The HSE06 hybrid functional reveals an indirect band gap of 5.090 eV, placing $\text{Sr}_2\text{PrSbO}_6$ in the deep-ultraviolet-transparent class of wide-gap oxides.

The elastic tensor yields polycrystalline VRH averages $B_H = 121.6$ GPa, $G_H = 60.9$ GPa, $E_H = 156.5$ GPa, and $\nu_H = 0.285$, with a Pugh ratio $B/G = 2.00$ classifying the oxide as ductile. BoltzTraP transport calculations on the GGA-PBE bands identify p-type conduction as the more favourable thermoelectric channel, with a Seebeck coefficient of 185–200 $\mu\text{V K}^{-1}$ at the optimal chemical potential ($\mu, -E_F = -0.141$ eV) and a power factor rising from 1.31 to 3.01 $\text{mW K}^{-2} \text{m}^{-1}$ over 300–1100 K. Combined with the phono3py lattice thermal conductivity, this yields a maximum figure of merit $ZT_{\text{max}} \approx 0.14$ at 1100 K. These parameters are relevant to heat management and self-powered sensing in the photodetector context.

First-principles-guided TCAD drift-diffusion simulations establish $\text{Sr}_2\text{PrSbO}_6$ as a solar-blind UV-C photodetector material with a hard optical cutoff at $\lambda_c = 243.6$ nm: the photoresponse vanishes for all $\lambda > 243.6$ nm, giving an extremely high, model-limited UV/visible rejection that follows directly from the abrupt vanishing of the absorption coefficient at the HSE06 band edge. Although the fundamental gap is indirect, operation at 220 nm accesses direct-allowed transitions ~ 0.55 eV above the gap onset, yielding a strong absorption coefficient $\alpha(220 \text{ nm}) = 5.02 \times 10^4 \text{ cm}^{-1}$ and a film absorption efficiency $\eta_{\text{abs}} \approx 53\%$ at a thickness of only 200 nm. Device performance is therefore governed not by optical absorption but by the photoconductive gain: the short carrier lifetime ($\tau = 1$ ns) relative to the carrier transit time ($t_{\text{tr}} \approx 50$ ns) limits the baseline responsivity to $R(220 \text{ nm}) = 2.82 \text{ mA/W}$ with a specific detectivity $D^* = 1.6 \times 10^9$ Jones for a 200 nm film at 10 V. Increasing the film thickness and bias, and most effectively the carrier lifetime, raises the projected responsivity to $\sim 44 \text{ mA/W}$ ($\tau = 10$ ns, $d = 5 \mu\text{m}$). These results identify $\text{Sr}_2\text{PrSbO}_6$ as a computationally screened candidate for deep solar-blind UV-C photodetection and identify carrier-lifetime improvement, rather than film thickness, as the primary route to device optimisation, motivating experimental synthesis and defect-controlled epitaxial growth for device characterisation.

CRediT Author Statement

Md. Mohiuddin: Conceptualisation, Formal analysis, Data curation, Writing – original draft, Visualisation. **Alamgir Kabir:**, Supervision, Writing – review and editing.

Declaration of Competing Interests

The authors declare no competing financial interests.

Data Availability

The raw VASP input and output files, Phonopy force constants, BoltzTraP2 input data, and all Python plotting scripts used to generate the figures in this work are available from the corresponding author upon reasonable request.

Supporting Information

The supplementary document contains Supplementary Methods, three static figures (Figs. S1–S3), five animated figures (Movies S4–S8), two tables (Tables S9–S10), and Supplementary Note 1 (Figs. S11–S14) presenting the thermoelectric transport properties of $\text{Sr}_2\text{PrSbO}_6$, followed by Supplementary References. Each animation is provided as a GIF file (for direct embedding) and an MP4 file (for video playback).

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